

LAB REPORT # 14

Title

CGPA Calculator in C++

Objective

To design a C++ program that calculates CGPA using:

- Student name
 - Subject details
 - Grade points
 - Credit hours
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Introduction

CGPA (Cumulative Grade Point Average) is used to measure academic performance. This program calculates CGPA by taking input of subjects, converting marks into grade points (manually entered), and using credit hours for weighted calculation.

Formula Used

$$\text{CGPA} = \frac{\sum (\text{Grade Point} \times \text{Credit Hours})}{\sum \text{Credit Hours}}$$

Algorithm

1. Start program
2. Input student name
3. Input number of subjects
4. For each subject:
 - Enter subject name
 - Enter marks
 - Enter grade point

- Enter credit hours
 - 5. Compute CGPA
 - 6. Display result
 - 7. End program
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Program Code

```
#include <iostream>
#include <string>
using namespace std;

int main() {

    string studentName, subjectName;
    int totalSubjects;
    float marks, gradePoint, creditHours;

    float totalCredits = 0, totalGradePoints = 0, cgpa;

    cout << "CGPA Calculator" << endl;

    cout << "Student Name: ";
    getline(cin, studentName);

    cout << "Total Subjects: ";
    cin >> totalSubjects;

    for (int i = 1; i <= totalSubjects; i++) {

        cout << "\nSubject " << i << endl;

        cout << "Name: ";
        cin >> subjectName;

        cout << "Marks: ";
        cin >> marks;

        cout << "Grade Point: ";
        cin >> gradePoint;

        cout << "Credit Hours: ";
        cin >> creditHours;

        totalCredits += creditHours;
        totalGradePoints += gradePoint * creditHours;
    }

    cgpa = totalGradePoints / totalCredits;

    cout << "\nStudent: " << studentName << endl;
    cout << "CGPA: " << cgpa << endl;
```

```
    return 0;  
}
```

Sample Output

Student: Ali
CGPA: 3.6

Conclusion

The program successfully calculates CGPA using grade points and credit hours with simple C++ logic.